

The sum of two independent gamma random variables

Asked 11 years, 8 months ago Modified 8 years, 5 months ago Viewed 46k times



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According to [the Wikipedia article on the Gamma distribution](#):

If $X \sim \text{Gamma}(a, \theta)$ and $Y \sim \text{Gamma}(b, \theta)$, where X and Y are independent random variables, then $X + Y \sim \text{Gamma}(a + b, \theta)$.



But I don't see any proof. Can anyone point me to its proof please?

Edit: Thanks to Zen a lot, and also I found the answer [as an example on the Wikipedia page about characteristic functions](#).



probability density-function gamma-distribution

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edited Mar 7, 2013 at 17:47



cardinal

27.3k

8

105

140

asked Mar 7, 2013 at 16:17



Dexter12

193

1

1

6

3 **Intuition:** Gamma(n) distributions arise as the sums of n independent Exponential distributions, whence it is immediate in this context that $X + Y$ will have a Gamma($a + b, \theta$) distribution provided a and b are both positive integers. – whuber ♦ Mar 7, 2013 at 16:54

3 Answers

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20

The proof is as follows: (1) Remember that the characteristic function of the sum of independent random variables is the product of their individual characteristic functions; (2) Get the characteristic function of a gamma random variable [here](#); (3) Do the simple algebra.

To get some intuition beyond this algebraic argument, check whuber's comment.



Note: The OP asked how to compute the characteristic function of a gamma random variable. If $X \sim \text{Exp}(\lambda)$, then (you can treat i as an ordinary constant, in this case)

$$\psi_X(t) = \mathbb{E}[e^{itX}] = \int_0^\infty e^{itx} \lambda e^{-\lambda x} dx = \frac{1}{1 - it/\lambda}.$$



Now use Huber's tip: If $Y \sim \text{Gamma}(k, \theta)$, then $Y = X_1 + \dots + X_k$, where the X_i 's are independent $\text{Exp}(\lambda = 1/\theta)$. Therefore, using property (1), we have

$$\psi_Y(t) = \left(\frac{1}{1 - it\theta} \right)^k.$$

Tip: you won't learn these things staring at the results and proofs: stay hungry, compute everything, try to find your own proofs. Even if you fail, your appreciation of somebody else's answer will be at a much higher level. And, yes, failing is OK: nobody is looking! The only way to learn mathematics is by fist fighting for each concept and result.

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edited Mar 7, 2013 at 18:26

answered Mar 7, 2013 at 16:48



Zen

25.1k

4

86

125

The referenced statement explicitly states "provided all X_i are independent." – whuber ♦ Mar 7, 2013 at 16:52

1 One thing I don't get though, is how did we arrive at the characteristic functions? – Dexter12 Mar 7, 2013 at 17:58

I'll add it to the answer. Take a look. – Zen Mar 7, 2013 at 18:02

Perhaps you can include a reference for the characteristic function of a $\Gamma(a, \theta)$ for non-integer values of a ? – Dilip Sarwate May 20, 2017 at 14:26



20

Here is an answer that does not need to use characteristic functions, but instead reinforces some ideas that have other uses in statistics. The density of the sum of independent random variables is the convolutions of the densities. So, taking $\theta = 1$ for ease of exposition, we have for $z > 0$,

$$\begin{aligned} f_{X+Y}(z) &= \int_0^z f_X(x) f_Y(z-x) dx \\ &= \int_0^z \frac{x^{a-1} e^{-x}}{\Gamma(a)} \frac{(z-x)^{b-1} e^{-(z-x)}}{\Gamma(b)} dx \\ &= e^{-z} \int_0^z \frac{x^{a-1} (z-x)^{b-1}}{\Gamma(a)\Gamma(b)} dx && \text{now substitute } x=zt \text{ and think} \\ &= e^{-z} z^{a+b-1} \int_0^1 \frac{t^{a-1} (1-t)^{b-1}}{\Gamma(a)\Gamma(b)} dt && \text{of Beta}(a,b) \text{ random variables} \\ &= \frac{e^{-z} z^{a+b-1}}{\Gamma(a+b)} \end{aligned}$$

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answered Mar 7, 2013 at 18:35



Dilip Sarwate

47.7k

4

124

235

3 (+1) It is ideal to have more than one way to prove everything. Maybe someone will post an answer considering the transformation $(X, Y) \mapsto (U, V) = (X + Y, X)$.
– Zen Mar 7, 2013 at 18:37

Can we similarly find the density of $X - Y$ in a closed form expression? I'm unable to simplify the integrals in that case. – pikachuchameleon Feb 26, 2016 at 20:33

@pikachuchameleon See [this answer](#) of mine. – Dilip Sarwate Feb 27, 2016 at 2:42



On a more heuristic level: If a and b are integers, the Gamma distribution is an Erlang distribution, and so X and Y describe the waiting times for respectively a and b occurrences in a Poisson process with rate θ . The two waiting times X and Y are

4

1. independent



2. sum up to a waiting time for $a + b$ occurrences



and the waiting time for $a + b$ occurrences is distributed $\text{Gamma}(a + b, \theta)$.



None of this is a mathematical proof, but it puts some flesh on the bones of the connection, and can be used if you want to flesh it out in a mathematical proof.

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answered Jun 15, 2016 at 17:36



Svein Olav Nyberg

171 1 4